

EX.NO: 10	PERFORM DATA CLEANING AND TRANSFORMATION CHALLENGE BY USING POWER BI
DATE:	

QUESTION - 1:

A retail company collects sales data from different sources such as Excel, SQL, and Web. The dataset contains missing values, duplicate records, and inconsistent formats. Perform data cleaning and transformation using Power BI to prepare the dataset for analysis and create meaningful visualizations.

AIM:

To clean, transform, and integrate data from multiple sources using **Microsoft Power BI** and generate a structured dataset for visualization and analysis.

ALGORITHM:

1. Load the dataset from Excel, SQL, and Web sources.
2. Identify missing values and duplicate records.
3. Remove duplicate rows from the dataset.
4. Handle missing values by replacing or removing them.
5. Convert data types into proper formats (Date, Currency).
6. Split or merge columns if required.
7. Create new calculated columns (Total Sales, Profit).
8. Filter unnecessary or invalid data.
9. Load the transformed data into Power BI.
10. Create dashboards for visualization.

PROGRAM:

Step 1: Import Libraries

```
import pandas as pd
```

```
import sqlite3
```

Step 2: Load Data from Excel

```
df_excel = pd.read_excel("Sales_Data.xlsx")
```

```
print("Excel Data:")
```

```
print(df_excel.head())
```

Step 3: Data Cleaning

Remove duplicates

```
df_excel = df_excel.drop_duplicates()
```

```
# Handle missing values
df_excel.fillna(0, inplace=True)

# Step 4: Create SQL Database Connection
conn = sqlite3.connect("sales.db")

# Step 5: Store Cleaned Data into SQL
df_excel.to_sql("sales", conn, if_exists="replace", index=False)
print("Data stored in SQL successfully!")

# Step 6: Read Data from SQL
query = "SELECT * FROM sales"
df_sql = pd.read_sql(query, conn)

print("Data from SQL:")
print(df_sql.head())

# Step 7: Data Transformation
# Create new columns
df_sql['Total Sales'] = df_sql['Quantity'] * df_sql['Price']
df_sql['Profit'] = df_sql['Total Sales'] - df_sql['Cost']

# Step 8: Save Output
df_sql.to_excel("Cleaned_Sales_Output.xlsx", index=False)
print("Output saved successfully!")

# Close connection
conn.close()
```

OUTPUT:

Cleaned Dataset Sample:

Product	Quantity	Price	Cost	Total Sales	Profit
Laptop	2	50000	40000	100000	20000
Phone	1	20000	15000	20000	5000

Mark Allocation		
Particulars	Marks Allotted	Marks Awarded
Program/Simulation	40	
Program Execution	30	
Result	20	
Viva Voce	10	
Total	100	

INFERENCE / RESULT:

- Power BI successfully connected to Excel, SQL, and Web data sources.
- Data cleaning removed duplicates and handled missing values effectively.
- Data transformation improved data quality and structure.
- Power Query simplified preprocessing tasks.
- Visual dashboards provided clear and interactive insights.
- The system supports better decision-making using clean data.